

DISTRIBUTION FUNCTION BASED LOSSES FOR PARAMETRIC PROBLEMS

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SUMMARY

We propose the assessment of losses in parametric estimation problems with a measure of the distance between the population and estimated distribution functions. Several classical results in estimation theory as well as the more recent minimax robustness theorems have analogues using the proposed losses. The class of loss functions considered is designed to reflect (unlike squared-error loss) in a flexible way the purposes behind the estimation.

1. Introduction

The theory of estimation has had great difficulty with the following question: What should be estimated when the assumed parametric model is incorrect (that is, does not contain the true distribution)? Only for special cases, such as the estimation of location of a symmetric distribution, have unique and sensible answers been proposed. For precisely this reason, Tukey (1962) suggested the symmetric location problem for initial study from the standpoint of robustness.

The suggestion has been made by Huber (1972) and Bickel and Lehmann (1975) that the chosen estimator determine the appropriate quantity to be estimated, i.e. $\theta_0 = \text{plim}_{n \rightarrow \infty} \hat{\theta}_n$.

This results, of course, in a destructive relativism with no means of objectively comparing different estimators in problems lacking a large number of symmetries. In location or scale problems, natural invariance principles make comparisons possible, but a truly general theory of estimation should extend beyond simple location and scale problems!

We herein adopt the view that the fundamental goal of (robust) estimation is the estimation of the *entire* distribution which has been sampled. A similar view is taken in Beran (1980a, 1980b) and Millar (1980, 1981) in which the discussion pertains to the issues treated here. Thus parameters will be of interest *only* inasmuch as they index the distributions in the model, and efficiency will be assessed in terms of the discrepancy between the true and fitted distribution functions, instead of points in the parameter space. Alternatively put, we shall view the distribution functions (probability measures) as themselves being infinite dimensional parameters, and assess errors via Hilbert space

norms.

2. Results in the correct model case

2.1 One-Dimensional Parameter Space

First, we consider the case of observing $X_1, X_2, \dots, X_n$ a random sample from F_{θ_0} , where $\Gamma = \{F_{\theta}, \theta \in \Omega\}$ is a parametric family of distribution functions. It is desired to estimate the true parametric value θ_0 based upon the n observations $X_1, X_2, \dots, X_n$. A typical measure of the loss due to an estimate $\hat{\theta}$ would be $w(\theta_0)(\hat{\theta} - \theta_0)^2$, weighted squared error loss. Unfortunately, this loss is not invariant under one-to-one reparametrizations of Ω (for instance, if we decide to estimate θ^3 instead of θ , for $\Omega \subseteq \mathbb{R}^1$). If the X_i are real-valued, an alternative loss would be $L(\hat{\theta}, \theta_0) = \int (F_{\hat{\theta}} - F_{\theta_0})^2 d\mu$, a weighted L^2 distance between $F_{\hat{\theta}}$ and F_{θ_0} , where μ is a nonnegative measure. Typical choices for μ will be such that

$$(2.1) \quad d\mu = dx(\text{Lebesgue}), d\mu = f_{\theta_0}^p(x)dx, \text{ or } d\mu = |x|^p dx,$$

for instance. The choice of μ in a specific problem would be determined by which aspects of the true distribution the experimenter deemed it most important to match. The second weight function in (2.1) would a) weight all points $x \in \mathbb{R}^1$ equally for $p = 0$, b) down-weight the importance of a good fit in regions of low probability for $p > 0$, giving a Cramer-von Mises type distance for $p = 1$, and c) give sizeable weight to fit in regions of low probability (typically the tails of the distribution) for $p < 0$. Thus, if estimation of extreme quantiles (or the survival function at extreme quantiles) is important, one would choose $p < 0$ to reflect the purpose of the parameter estimation. Conversely, $p > 0$ would be appropriate if a good fit to the "middle part" of the distribution were desired.

Typically, finite sample computations of the risk (expected loss) of an estimator are, for practical purposes, intractable. Concrete expressions are possible for the uniform and gamma distributions, for suitable weight functions. However, we prefer to concentrate on the conceptually simpler asymptotic case.

Let $X_1, X_2, \dots$ be iid from F_{θ} .

Definition

If $Z_n = n \int (F_{\theta_n} - F_{\theta})^2 d\mu \xrightarrow{L} Z$, where $\{\theta_n\}$ is a sequence of random variables considered as estimators of θ , and $0 < E_{\theta}[Z] < \infty$, we define $E_{\theta}[Z]$ as the first-order asymptotic risk of the estimator (sequence) θ_n , at the point θ .

Under the following regularity conditions we may express $E_\theta [Z]$ as a simple function of the asymptotic variance of θ_n . E1 requires that θ_n have, after a normalizing factor of $\sqrt{n}$, a finite asymptotic variance. F1 imposes a smoothness condition on the model Γ .

E1: As $n \rightarrow \infty$, $\sqrt{n}(\theta_n - \theta) \xrightarrow{L} Y$, where $E[Y^2] = \sigma^2 < \infty$.

F1: There exists a function $\eta_\theta \in L^2(\mu)$ such that

$$\int (F_{\theta_n} - F_\theta - (\theta_n - \theta) \eta_\theta)^2 d\mu = o(|\theta_n - \theta|^2) \text{ as } \theta_n \rightarrow \theta,$$

$$\theta_n \in \Omega \text{ for all } n, \theta \in \Omega.$$

Typically we have $\eta_\theta(x) = \frac{\partial F_\theta(x)}{\partial \theta}$, and our condition F1 is just quadratic differentiability of F_θ . In fact, for $\int |d\mu| < \infty$ it follows as a consequence of ordinary quadratic differentiability as used in Lecam (1970) (see Pollard (1980, p. 47)).

Theorem 1

Let $X_1, X_2, \dots$ be iid from F_θ , and let θ_n be a function of $X_1, \dots, X_n$ such that E1 holds, and Γ be such that F1 holds. Then $n \int (F_{\theta_n} - F_\theta)^2 d\mu \xrightarrow{L} Z$, and the asymptotic risk of θ_n is given by $E_\theta [Z] = \sigma^2 \int \eta_\theta^2 d\mu$.

Hence, in the typical one-dimensional case the comparison between estimators based upon $E_\theta [Z]$ is asymptotically equivalent to a comparison based upon their variances. For small samples, where the variance criterion may well seem less reasonable, the risk has a concrete physical interpretation as a weighted L^2 distance between the estimated and true distribution functions F_{θ_n} and F_θ , the weight choice reflecting the portions of the distribution where a good fit is desired.

Another good property of this measure is that it is completely invariant, even for small samples, under one-to-one reparametrizations of Γ . This property is, of course, not enjoyed by the ordinary mean-squared-error or variance based measures. In addition, it is no more difficult to compute in the asymptotic case than mean-squared-error.

Under further regularity of Γ and θ_n , we may obtain an asymptotic expansion of the risk up to and including terms which are $O(\frac{1}{n})$. With this in mind we introduce the following conditions, taking the true value of the parameter θ to be 0 without loss of generality and writing $F = F_0$ for notational convenience:

E2: As $n \rightarrow \infty$, the moments of θ_n satisfy

$$(i) \quad E[\theta_n^2] = \frac{a_1}{n} + \frac{a_2}{n^{3/2}} + \frac{a_3}{n^2} + o(n^{-2})$$

$$(ii) \quad E[\theta_n^3] = \frac{b_1}{n^{3/2}} + \frac{b_2}{n^2} + o(n^{-2})$$

$$(iii) \quad E[\theta_n^4] = \frac{c_1}{n^2} + o(n^{-2})$$

F2: There exist functions η , ξ and ϕ , all in $L^2(\mu)$, such that

$$\int (F_{\theta_n} - F - \theta_n \eta - \theta_n^2 \xi - \theta_n^3 \phi)^2 d\mu = o(\theta_n^6) \text{ as } \theta_n \rightarrow 0.$$

Condition E2 is just a typical moment expansion for an estimator which is fairly "regular". In most cases we have $\eta = \frac{\partial F_\theta}{\partial \theta}$, $\xi = \frac{1}{2} \frac{\partial^2 F_\theta}{\partial \theta^2}$ and $\phi = \frac{1}{6} \frac{\partial^3 F_\theta}{\partial \theta^3}$, and thus condition F2 may be termed "quadratic thrice-differentiability of F ". Quadratic differentiability of orders other than one or three has a definition analogous to F1 and F2. We are now able to develop a second order expansion of the risk.

Theorem 2

Let $X_1, X_2, \dots$ be iid from $F_0 = F$, and let θ_n be a function of $X_1, \dots, X_n$ such that E2 holds, and P be such that F2 holds. If in addition $E[\theta_n^6] = o(\frac{1}{n^2})$ as $n \rightarrow \infty$, then

$$\begin{aligned} E[n \int (F_{\theta_n} - F)^2 d\mu] = \\ \cdot a_1 \int \eta^2 d\mu + [a_2 \int \eta^2 d\mu + 2b_1 \int \eta \xi d\mu] \frac{1}{n^{1/2}} \\ + [a_3 \int \eta^2 d\mu + 2b_2 \int \eta \xi d\mu + c_1 \int (\xi^2 + 2\eta\theta) d\mu] \frac{1}{n} + o(n^{-1}). \end{aligned}$$

The proof is straightforward and hence omitted.

If we consider the special case of θ a location parameter and μ Lebesgue measure, we obtain, if f has support R^1 ,

$$\begin{aligned} E[n \int (F_{\theta_n} - F)^2 d\mu] = a_1 \int_{-\infty}^{\infty} f^2(x) dx + \frac{a_2}{n^{1/2}} \int_{-\infty}^{\infty} f^2(x) dx \\ + \frac{1}{n} (a_3 \int_{-\infty}^{\infty} f^2(x) dx + \frac{7c_1}{12} \int_{-\infty}^{\infty} (f'(x))^2 dx) + o(\frac{1}{n}). \end{aligned}$$

Hence, the skewness of θ_n does not affect the asymptotic risk up to and including terms of order $O(\frac{1}{n})$. All terms present in the expansion above are in fact due to the variance of

θ_n , except for the kurtosis (c_1) contribution to the $O(\frac{1}{n})$ term. Thus, the analysis based upon the loss suggested here agrees with a mean-square error based analysis up to terms of $O(\frac{1}{n})$. For the standard normal parent, the expansion becomes

$$\frac{a_1}{2\sqrt{\pi}} + \frac{a_2}{2\sqrt{(n\pi)}} + \frac{1}{n} \left(\frac{a_3}{2\sqrt{\pi}} + \frac{7c_1}{48\sqrt{\pi}} \right) + o\left(\frac{1}{n}\right).$$

The author has not yet been able to characterize the class of "second-order optimum" estimators.

2.2 Multi-Dimensional Parameter Space

The major benefits of assessing losses with L^2 norms on distribution functions accrue in the multi-dimensional parameter case. Here, classical methods of estimator comparison are based upon the (asymptotic) variance-covariance matrix of the parameter estimator. Notationally, we let $\Gamma = \{F_\theta, \theta \in \Omega\}$, where Ω is a subset of R^K , and Γ is again a parametric family of distribution functions. We define the asymptotic risk of an estimator sequence θ_n as in Section 2.1, except that θ_n is now R^K -valued. The following extensions of regularity conditions E1 and F1 to the multiparameter case are necessary: (We take $\theta = \underline{0}$, a K -vector of zeroes.)

EM: As $n \rightarrow \infty$, $\sqrt{n}\theta_n \xrightarrow{L} Y$, where $E[Y] = \underline{0}$ and $\text{Var}(Y) = \Sigma = \{\sigma_{ij}\}$

FM: There exist functions $\eta_1, \eta_2, \dots, \eta_K$ in $L^2(\mu)$ such that

$$\int (F_{\theta_n} - F - \sum_{j=1}^K \theta_{jn} \eta_j)^2 d\mu = o(|\theta_n|^2), \text{ where}$$

$$\theta'_n = (\theta_{1n}, \theta_{2n}, \dots, \theta_{Kn}).$$

In typical applications Σ , the asymptotic variance-covariance matrix of $\sqrt{n}\theta_n$, is used as the basis for classical efficiency definitions through trace (Σ) or $\det(\Sigma)$. Usually

$\eta_n(x) = \frac{\partial F_\theta(x)}{\partial \theta_j} \bigg|_{\theta=\underline{0}}$. Once again, FM follows as a consequence of quadratic differentiability.

Theorem 3

Let $X_1, X_2, \dots$ be iid from $F_0=F$, and let θ_n be a function of $X_1, \dots, X_n$ such that EM holds, and Γ be such that FM holds. Then $\int (F_{\theta_n} - F)^2 d\mu \xrightarrow{L} Z$, and the asymp-

total risk is given by

$$E[Z] = \sum_{j=1}^K \sum_{\ell=1}^K \sigma_{j\ell} \int \eta_j \eta_{\ell} d\mu.$$

Notes:

- (1) The proof proceeds as that for the one-dimensional case, by simple algebra.
- (2) The risk is a weighted average of quadratic forms in Σ . That is setting $\eta'(x) = (\eta_1(x), \eta_2(x), \dots, \eta_K(x))$, we have $E[Z] = \int \eta' \Sigma \eta d\mu$. The weights are provided by the measure μ .
- (3) For this note and those following, let Σ_1 and Σ_2 be the asymptotic variance-covariance matrices of two vector-valued estimator sequences $\sqrt{n}\theta_n^1$ and $\sqrt{n}\theta_n^2$, in a situation where EM and FM are satisfied, and R_1 and R_2 be their respective risks. Then, if $\Sigma_1 - \Sigma_2$ is positive semidefinite, we have $R_1 \geq R_2$. Further, if $\Sigma_1 - \Sigma_2$ is positive definite, then $R_1 > R_2$, if μ assigns nonzero mass to the real line. Hence, in those situations where the classical theory yields an unambiguous answer, the current method agrees.
- (4) If $\Sigma_1 - \Sigma_2$ is neither positive semidefinite or negative semidefinite, then there exists a choice of μ such that $R_1 > R_2$ and a choice of μ such that $R_1 < R_2$. In such cases the choice of μ , reflecting the relative importance of accurate estimation over different parts of the distribution, determines which estimator is preferable.
- (5) There is no need for asymptotic normality of $\sqrt{n}\theta_n^1$ or $\sqrt{n}\theta_n^2$ to justify the comparison.
- (6) Overparametrization of Γ , which typically results in $|\Sigma| = 0$, causes no problems for the theory.
- (7) Super-efficiency of estimation of one element θ_i in the parameter vector simply results in comparisons being based upon the other elements of θ .

Example

Let X_i be iid $N(\theta_1, \theta_2^2)$ for $-\infty < \theta_1 < \infty$, $\theta_2 > 0$. Hence $\theta' = (\theta_1, \theta_2)$. Consider the class of losses given by $d\mu = f_0^p(x)dx$, where f_0 is the standard normal density.

Letting

$$\begin{bmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{bmatrix}$$

be the asymptotic variance-covariance of $\sqrt{n}(\theta_n - \theta)$, we find the risks to be as follows (up to irrelevant multiplicative constants): i) $\sigma_{11} + 1/3\sigma_{22}$ for $p = 1$, ii) $\sigma_{11} + 1/2\sigma_{22}$ for $p = 0$, and iii) $\sigma_{11} + \sigma_{22}$ for $p = -1$. Hence, larger weight given by μ to the tails places a greater premium upon precise estimation of the scale parameter in this case.

3. Results in incorrect model cases

3.1 Fixed Parent Distribution

We modify the notational set-up of Section 2 as follows: $X_1, X_2, \dots$ are iid G , with

$G \notin \Gamma = \{F_\theta, \theta \in \Omega\}$. Clearly, if μ is a finite measure, we have

$$(3.1) \quad \lim_{n \rightarrow \infty} \int (F_{\theta_n} - G)^2 d\mu = \int (F_\theta - G)^2 d\mu,$$

where the limit is in whatever sense $\hat{\theta}_n \rightarrow \theta$, if $F_\theta(x)$ is a continuous function of θ at all x but a set of μ -measure 0. If we assume further that the right-hand side of (3.1) attains its infimum at a unique value of θ , denoted $\theta(G)$, we observe that a minimum distance estimator sequence $\{\theta_n\}$ is defined by

$$(3.2) \quad \int (F_{\theta_n} - G_n)^2 d\mu = \inf_{\theta \in \Omega} \int (F_\theta - G_n)^2 d\mu,$$

where G_n is the empirical distribution function. This sequence will be such that the expected loss converges to $\int (F_{\theta(G)} - G)^2 d\mu$, the minimum possible value. If the infimum either is not attained or is attained at more than one value, a device such as that of Parr and Schucany (1980, Definition 1) may be employed to deal with the problem. We shall assume throughout this subsection that $\theta(G_n)$ is uniquely defined with probability one for sufficiently large n to avoid technical complications which contribute essentially no new insight. The following theorem, while natural and simple to prove, states the superiority of minimum distance estimators in this context.

Theorem 4

Let $\{\theta_n\}$ be an estimator sequence of functional type based upon G_n , that is, assume that there exists a functional T such that $T[G_n] = \theta_n$ with probability one. Assume that T is a continuous functional with respect to the sup-metric. If H is a class of distributions containing a Kolmogorov ball around a distribution G , then

$$(3.3) \quad \lim_{n \rightarrow \infty} \int (F_{\theta_n} - H)^2 d\mu = \int (F_{\theta(H)} - H)^2 d\mu$$

for all $H \in H$ with probability one iff, with probability one, for n sufficiently large, $\theta_n = \theta(G_n)$, i.e. θ_n is a minimum distance estimator. For the sup-metric (and Kolmogorov ball) we could substitute any other metric in which the empirical distribution function converges (and an associated neighbourhood).

Since the right-hand side of (3.3) is the minimum value achievable for the risk, the

result says that any estimator with optimal zero-order asymptotic risk in a full neighbourhood of a distribution is for large samples *exactly* equivalent to a minimum distance estimator. If the neighbourhood is not so rich (perhaps finite-dimensional) or the estimator is not continuous (perhaps highly adaptive) this equivalence might fail.

If θ_n is an estimating sequence optimizing this zero-order asymptotic risk and F_θ is twice quadratically differentiable with notation for the "derivatives" as in Section 2, we may obtain an expansion for the loss as

$$(3.4) \quad \int (F_{\theta_n} - G)^2 d\mu \\ = \int (F_{\theta(G)} - G)^2 d\mu + (\theta_n - \theta(G))^2 \left\{ \int \eta^2 d\mu + \int \phi(F_{\theta(G)} - G) d\mu + o(|\theta_n - \theta(G)|^2) \right\}$$

as $n \rightarrow \infty$. Hence the loss becomes (essentially) ordinary quadratic loss, when comparing estimators which are zero-order optimal.

3.2 Parent Distribution Shrinkage Towards the Model

Let $X_{in}, i = 1, \dots, n$ be iid with distribution

$$(3.5) \quad G^n = F + \frac{\gamma}{\sqrt{n}} (H - F)$$

for $n = 1, 2, \dots$ where $\gamma > 0$ and H is a distribution unknown to the statistician. Thus, at sample size n , the data come from a mixture of two distributions, with proportion $1 - \frac{\gamma}{\sqrt{n}}$ of the mixture coming from a distribution in the model, and the other $\frac{\gamma}{\sqrt{n}}$ representing arbitrary contamination. We shall require that assumption F1 hold at $\theta = 0$, and assume that the estimating sequence is of functional type and differentiable at F with respect to the sup-norm in the sense of the following definition.

Definition

A functional T from the space of distribution functions into R^1 is differentiable at F with respect to the sup-norm iff for any sequence $H_1, H_2, \dots$ of distribution functions such that $\sup_x |H_n(x) - F(x)| \rightarrow 0$,

$$(3.6) \quad \lim_{n \rightarrow \infty} \frac{|T[H_n] - T[F] - D(H_n)|}{\sup_x |H_n(x) - F(x)|} = 0,$$

where $D(K) = \int \psi(x) dK(x)$, for all distribution functions K and some function ψ which is continuous almost everywhere.

Note: Since we do **not** require that ψ be bounded and continuous, "differentiability" (in our sense) of T does not imply continuity of T — see for instance the mean functional.

Lemma

Let T be differentiable at F with respect to the sup-norm. Then, if G_n is the empirical distribution function based upon X_{in} , $i = 1, \dots, n$ we have

$$(3.7) \quad \sqrt{n}(T[G_n] - T[F]) \xrightarrow{L} N(\gamma \int \psi(x) dH(x), \int \psi^2(x) dF(x)),$$

as $n \rightarrow \infty$, if $0 < \int \psi^2(x) dF(x) < \infty$ and $\int \psi^2(x) dH(x) < \infty$

Proof

The result follows immediately by the definition above and the Lindeberg-Levy version of the central limit theorem for triangular arrays.

Theorem

Let T be a functional differentiable at F with respect to the sup-norm. Let $T[F] = 0$ w.l.o.g., and write $F_0 = F$. Then, $\sqrt{n}(T[G_n] - F)^2 d\mu \xrightarrow{L} Z$, where

$$(3.8) \quad E[Z] = \int \psi^2(x) dF(x) + \gamma^2 (\int \psi(x) dH(x))^2,$$

with ψ defined by (3.6).

For fixed ψ , this quantity is clearly maximized over choice of H as

$$(3.9) \quad \int \psi^2(x) dF(x) + \gamma^2 b^2,$$

where $b = \sup_x |\psi(x)|$. Hence our problem reduces to that of Hampel (1974, p. 391) under his regularity conditions, since b is Hampel's gross-error sensitivity. Hence, the estimator T which is minimax with respect to the statistical game in which nature picks the contaminating distribution H , the statistician picks T , and the loss is given by (3.8) must be of the form

$$(3.10) \quad T[G] = T[F] + \int \psi(x) dG(x) + o(\|G - F\|_\infty)$$

where

$$\psi(x) = c \min(|\frac{d}{d\theta} \ln f_\theta(x) - \alpha|, d) \operatorname{sign} \{ \frac{d}{d\theta} \ln f_\theta(x) - \alpha \}$$

for some $d > 0$, $c \neq 0$, and real α .

Some might argue that the loss used in this section, $n \int (F_{\theta_n} - F)^2 d\mu$, is inappropriate, and that $n \int (F_{\theta_n} - G^n)^2 d\mu$ is to be preferred. This alternate loss demands that the estimator be penalized when G^n is far from Γ , (that is, for badly contaminated data or an otherwise poor model), regardless of how well the estimation is done. Our loss instead asks the estimator to search for the parameter corresponding to the portion of the data from the model, and penalizes based upon how well this goal is accomplished. These competing viewpoints are termed empiricism and idealism by Beran (1980b), and also appear in the discussion of Stigler (1977) pertaining to bias.

Beran (1980a) considers losses of the form $w(d(F_{T_n}^n, G^{n,n}))$, where w is a non-negative monotone increasing function from $[0, \infty)$ into R^1 , $F_{T_n}^n$ and $G^{n,n}$ are n -fold product measures with marginals F_{T_n} and G^n respectively, and d is Hellinger distance, i.e.

$$d(H, K) = \int (h^{1/2}(x) - k^{1/2}(x))^2 dx,$$

where h and k are the densities of H and K respectively. At sample size n , a random sample is taken from G^n , where G^n is in (a subset of) a Hellinger ball of radius $O(n^{-1/2})$ centered about F_0 . If w is a bounded function, Beran shows that asymptotic efficiency and robustness are compatible in his sense, i.e. that an estimator fully efficient at the model is asymptotically minimax in the sense of his Theorem 1. Hence, for bounded w his set-up evaluates the notorious sample mean as robust in normal models. This is not the case, however, for unbounded w , when he finds the sample mean to not be asymptotically minimax. It should be noted that contamination neighbourhoods which are contained in Hellinger balls do not allow for contamination which is singular with respect to the model (for instance, rounding errors). Hence, as recognized by Beran (1980a, p. 1257), the contamination neighbourhoods are quite small and hence the robustness properties (asymptotic minimaxity over his shrinking contamination neighbourhoods) are thus rather limited. In a later paper (Beran (1981)), he studies estimators under a quadratic loss function, where the "parameter" to be estimated at sample size n is (almost) that value in the parameter space minimizing Hellinger distance from the sampled distribution to the model. Hence, he adopts the viewpoint he has termed empiricism. In such a set-up he derives estimators which are asymptotically efficient at the model and robust against a full Hellinger ball contamination model. Beran (1980b) studies the extension of these ideas to independent non-identically distributed data, characterizing asymptotically minimax (robust) parameter estimators with a different and more general choice of neighbourhoods for the distributions sampled (one possibility is $L^2(\mu)$ balls) and the "parameter" to be estimated defined through the same ideas as in Beran (1980a), adapted to the different choice of neighbourhoods in the latter paper. Asymptotically minimax robust

estimators as derived here are not in general asymptotically efficient, even at the model.

Millar (1981) uses a loss function *closely related* to $\int (F_{\theta_n} - G^n)^2 d\mu$, i.e. the empiricist choice for an appropriate loss. In such a context, he finds the natural minimum $L^2(\mu)$ distance estimator to be asymptotically minimax (robust), when the sampled distributions G^n are in $L^2(\mu)$ balls of radius $O(n^{-1/2})$ at stage n . He extends these ideas to the regression problem in Millar (1980).

Our loss is (as mentioned above) distinct from those of Millar (1980) and Beran (1980a) in that it does not penalize the estimator just because the data are badly contaminated (i.e. the model is poor), but only if the estimator behaves badly in choosing a poor approximation in Γ to F_{θ_0} .

4. Summary

We have proposed the assessment of loss in parametric estimation with a measure of distance between two distribution functions. The resultant losses give some results comparable to classical squared-error loss due to their locally quadratic character, but give an intuitively reasonable method of dealing with the problem of what should be estimated when the model is incorrect.

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